

Abstract Rendering: Certified Rendering Under 3D Semantic Uncertainty

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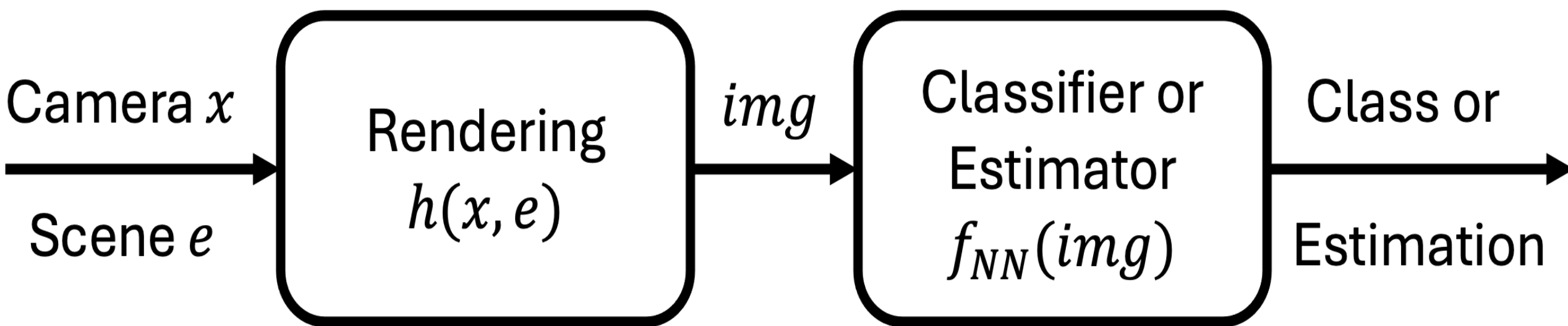
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INTRODUCTION

Problem:

Consider a typical visual autonomy pipeline:



Model robustness is usually studied under small, pixel-level perturbations of images (img). Can we instead verify robustness of a vision pipeline under **large semantic variations** in camera pose x and scene e ?

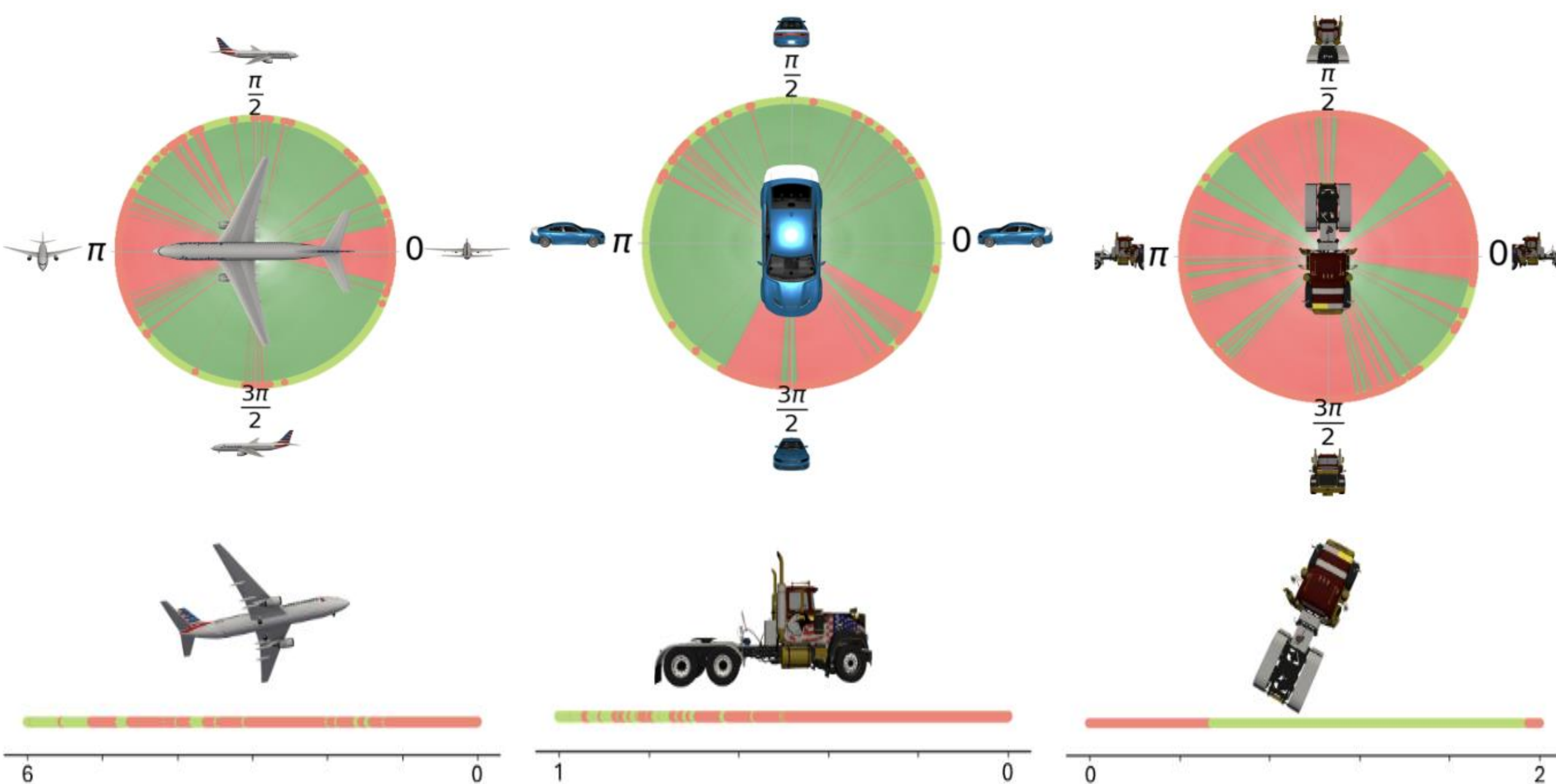
Abstract Rendering enables such analysis. For example, we can verify that “Over a 20-meter camera motion in the XYZ axes, a neural position estimator for the aircraft yields at most a 3-meter error when the target is 60 meters away.”

Methodology:

1. Treat the rendering and downstream network as composition of various operations.
2. Compute linear relational approximation for each operation.
3. Propagate bounds from inputs to outputs step by step through relational approximations.

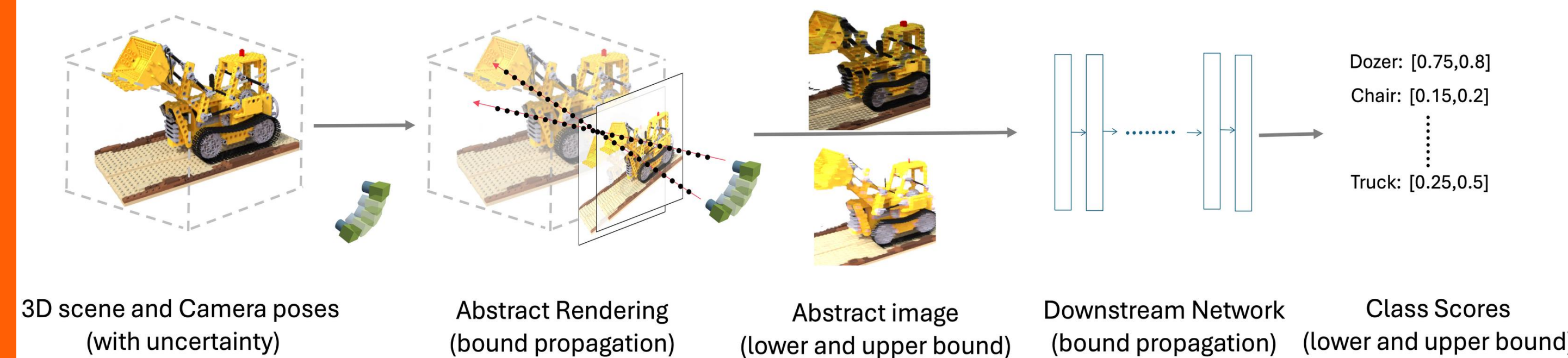
Application:

Verify the ranges of camera movement for which f_{NN} is certified to work—and identify the ranges where it may fail. 1) Classifier under 360° view; 2) Pose estimator under camera translation.

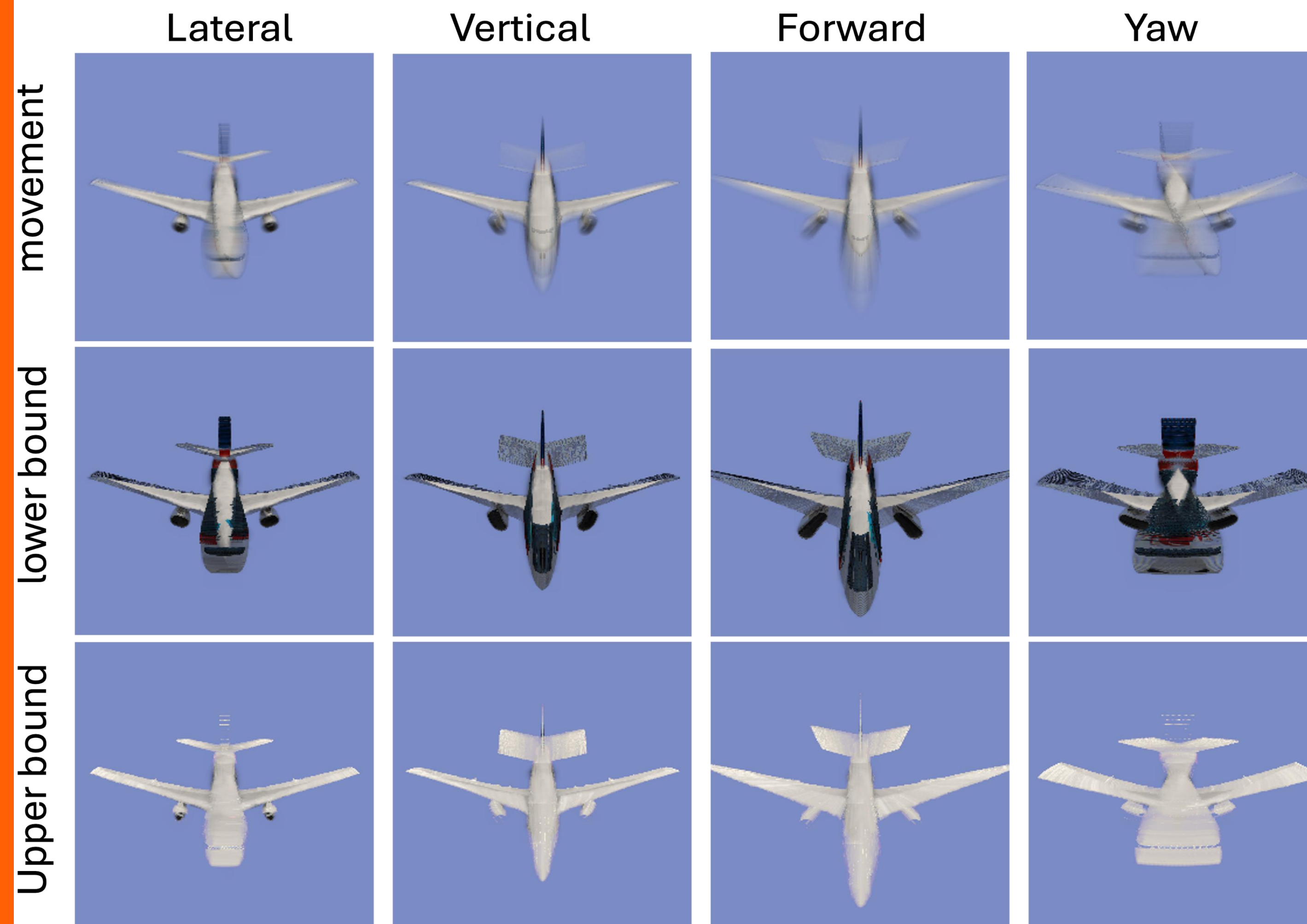


ABSTRACT RENDERING

Abstract Rendering + NN Verification:



Abstract Images:



- **First** framework for computing abstract images of scenes represented by Gaussian Splats and NeRFs under camera pose or scene uncertainty.
- **Novel linear relational approximations** of three rendering-specific operations: matrix inversion, sorting-based aggregation, cumulative product summation.
- By integrating Abstract Rendering with CROWN, we have enabled certification of visual tasks with respect to semantic variations in 3D environments.

EXPERIMENTS

Metrics: **Mean Pixel Gap (MPG)**, and **Max Pixel Gap (XPG)**:

$$MPG = \frac{1}{|WH|} \sum_{j \in WH} \|\tilde{c}_j - c_j\|, XPG = \max_{j \in WH} \|\tilde{c}_j - c_j\|$$

Table: Abstract Rendering results along with empirical bounds

Scene	CPR	Dim	Res	GS			NeRF			Empirical	
				Rt	MPG	XPG	Rt	MPG	XPG	MPG	XPG
Lego	0.1 (rad)	yaw	80×80	22	0.51	1.73	25	0.40	1.73	0.22	1.57
Chair	0.1 (rad)	yaw	50×50	17	0.46	1.73	20	0.40	1.73	0.40	1.72
Drums	0.1 (m)	x	50×50	5.3	0.19	1.62	5.8	0.16	1.55	0.13	1.54
PineTree	2 (m)	x	72×72	4.1	0.27	1.73	4.9	0.23	1.73	0.06	1.37
PineTree	10 (m)	x	72×72	24	0.47	1.73	28	0.36	1.73	0.18	1.37
Airport	0.027 (rad)	roll	160×160	27	0.56	1.69	35	0.42	1.63	0.21	1.38
Airport	0.03 (rad)	roll	160×160	30	0.59	1.70	43	0.51	1.68	0.22	1.41
Garden	0.5 (m)	x	100×100	20	0.53	1.59	11	0.37	1.53	0.34	1.21
Airplane	2 (m)	x	80×80	16	0.41	1.68	19	0.33	1.61	0.15	1.39

Abstract Images under Scene Variations:

- 1) Gaussian scene with variations on two roadside trees; 2) NeRF scene with hue or saturation shift.

